
Predicting FIFA World Cup 2026 Outcomes Using Machine Learning

Group Project Report

CS 445: Machine Learning

Portland State University

Spring 2026

Veniamin Velikoretskikh and Pete Treemeth

Professor Anthony Rhodes

June 9, 2026

Team Member Contributions:

Veniamin Velikoretskikh: Veniamin was responsible for the coding portion of the project, including building the machine learning pipeline, training the models, generating the evaluation metrics, and producing the plots used to compare the results.

Pete Treemeth: Pete was responsible for the written documentation, including organizing the report, explaining the dataset and methodology, interpreting the results, and editing the final submission.

Abstract:

This report compares machine learning algorithms for predicting FIFA World Cup 2026 outcome labels from team-level soccer statistics. The dataset includes features related to ranking, recent form, scoring, defense, player quality, and match context. Logistic Regression was used as a linear baseline, while an SVM with an RBF kernel was used to test whether a nonlinear classifier would perform better. Both models were evaluated with and without Principal Component Analysis. The SVM without PCA produced the strongest validation results, reaching an accuracy of 0.6600 and an F1-score of 0.6344. PCA did not improve performance for either model, suggesting that the original feature set was more useful than the reduced representation created by PCA.

1 Introduction

Predicting soccer outcomes is difficult because team performance depends on many factors, including ranking, recent form, scoring ability, defensive strength, player quality, travel distance, and host advantage. Since no single feature is likely to explain the result by itself, this problem is a reasonable fit for supervised machine learning.

The goal of this project is to compare models for predicting FIFA World Cup 2026 outcome labels using team-level statistics. The target variable is a binary winner label, so the task is handled as binary classification. We compare Logistic Regression as a linear baseline against an SVM with an RBF kernel as a nonlinear model. Each model is also tested with and without PCA to determine whether dimensionality reduction improves performance or removes useful information.

2 Background and Related Work

Machine learning is commonly used in sports analytics for player evaluation, team ranking, match prediction, and tournament forecasting. These tasks are challenging because sports outcomes depend on both measurable statistics and unpredictable events. Rankings, recent wins, scoring averages, and player ratings can provide useful signals, but they do not capture everything that affects a match, such as injuries, coaching decisions, weather, pressure, or random in-game events.

Logistic Regression is often used as a baseline for binary classification because it is simple and efficient. However, as a linear model, it may be limited when the relationship between features and labels is more complex. Support Vector Machines (SVM) can address this limitation by using kernels to form nonlinear decision boundaries (Cortes and Vapnik, 1995). In this project, the RBF kernel was used because soccer outcomes may depend on combinations of features rather than isolated statistics.

Principal Component Analysis (PCA) was included as a dimensionality reduction experiment (Jolliffe, 2002). It can reduce redundancy in the feature space, but because PCA does not use the target label, it does not necessarily preserve the information most useful for classification. For that reason, PCA was tested rather than assumed to be helpful.

3 Dataset Description

The dataset comes from a FIFA World Cup 2026 prediction dataset and contains team-level soccer statistics (Rambe, 2026). The labeled training file contains 1000 examples, while the separate test file contains 250 examples without true winner labels. Since the test file cannot be scored, model performance was evaluated using an 80/20 train-validation split from the labeled training data.

The target variable is the winner column. Before training, the columns `team_name`, `country_code`, and `confederation` were removed. The first two were removed because they function more like identifiers than general performance features. The confederation column was removed to keep the current feature matrix fully numeric. After removing these columns and the target label, the model used 22 input features. The final split contained 800 training examples and 200 validation examples, with stratification used to preserve the class balance.

4 Methodology

The project was implemented in Python using Pandas, NumPy, Matplotlib, and scikit-learn, a standard machine learning library for Python (Pedregosa et al., 2011). Pandas was used for loading and preparing the CSV files, scikit-learn was used for preprocessing, model training, PCA, and evaluation, and Matplotlib was used to generate the result plots.

The implementation used scikit-learn pipelines so that preprocessing and model training could be applied consistently. Each model first scaled the input features using `StandardScaler`, optionally applied PCA, and then trained the classifier. Four configurations were evaluated:

1. Logistic Regression without PCA
2. Logistic Regression with PCA
3. SVM with RBF kernel without PCA
4. SVM with RBF kernel with PCA

4.1 Data Processing

The program loads `train.csv` and `test.csv`. Since `test.csv` does not include true labels, it was not used for accuracy, precision, recall, or F1-score. Instead, the labeled training data was split into training and validation sets.

The preprocessing step separates the winner column as the target variable and keeps the remaining numerical columns as the input features. The data is then split using an 80/20 stratified split. Feature scaling is applied with `StandardScaler` because the dataset contains variables with different ranges, such as FIFA points, market value, possession percentage, and travel distance. Standardization prevents larger-scale variables from dominating the models.

4.2 Model Training

The Logistic Regression model was trained as the baseline. Since it is linear, its performance gives a useful reference point for judging whether the SVM's nonlinear decision boundary provides an improvement. The SVM model used an RBF kernel because the outcome label may depend on nonlinear combinations of team statistics.

The PCA versions of the models used the same classifiers, but PCA was inserted between scaling and model training. PCA was set to preserve 95% of the variance in the input features. This allowed the reduced feature representation to keep most of the variation in the dataset while still testing whether dimensionality reduction improved the classifiers.

5 Evaluation

The models were evaluated using the validation set created from the labeled training data. The original training file contained 1000 labeled examples, which were split into 800 training examples and 200 validation examples. The validation set was used to estimate how well each model would perform on unseen data.

The main evaluation metrics were accuracy, precision, recall, and F1-score. Accuracy measures the overall percentage of correct predictions. Precision measures how many teams predicted as winners were actually winners. Recall measures how many actual winners were correctly identified by the model. F1-score combines precision and recall, which makes it useful when comparing models that may trade off between false positives and false negatives.

Confusion matrices were also generated for each model. These show the actual types of mistakes each classifier made. For this project, the confusion matrices helped show whether a model was better at identifying winners, not-winners, or whether it made similar errors across both classes.

6 Results

The results showed that the SVM without PCA performed the best overall. It achieved a validation accuracy of 0.6600 and an F1-score of 0.6344. Logistic Regression without PCA was close behind, with a validation accuracy of 0.6500 and an F1-score of 0.6277. The difference between the two models was small, but the SVM had the strongest overall performance.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression without PCA	0.6500	0.6344	0.6211	0.6277
Logistic Regression with PCA	0.6200	0.6022	0.5895	0.5957
SVM with RBF kernel without PCA	0.6600	0.6484	0.6211	0.6344
SVM with RBF kernel with PCA	0.6300	0.6154	0.5895	0.6022

Table 1. Validation performance across all model configurations

The models without PCA performed better than the models with PCA. Logistic Regression dropped from an F1-score of 0.6277 without PCA to 0.5957 with PCA. The SVM also dropped from an F1-score of 0.6344 without PCA to 0.6022 with PCA. This suggests that PCA did not help with this dataset. Although PCA preserved most of the variance, it may have compressed or removed information that was useful for predicting the winner label.

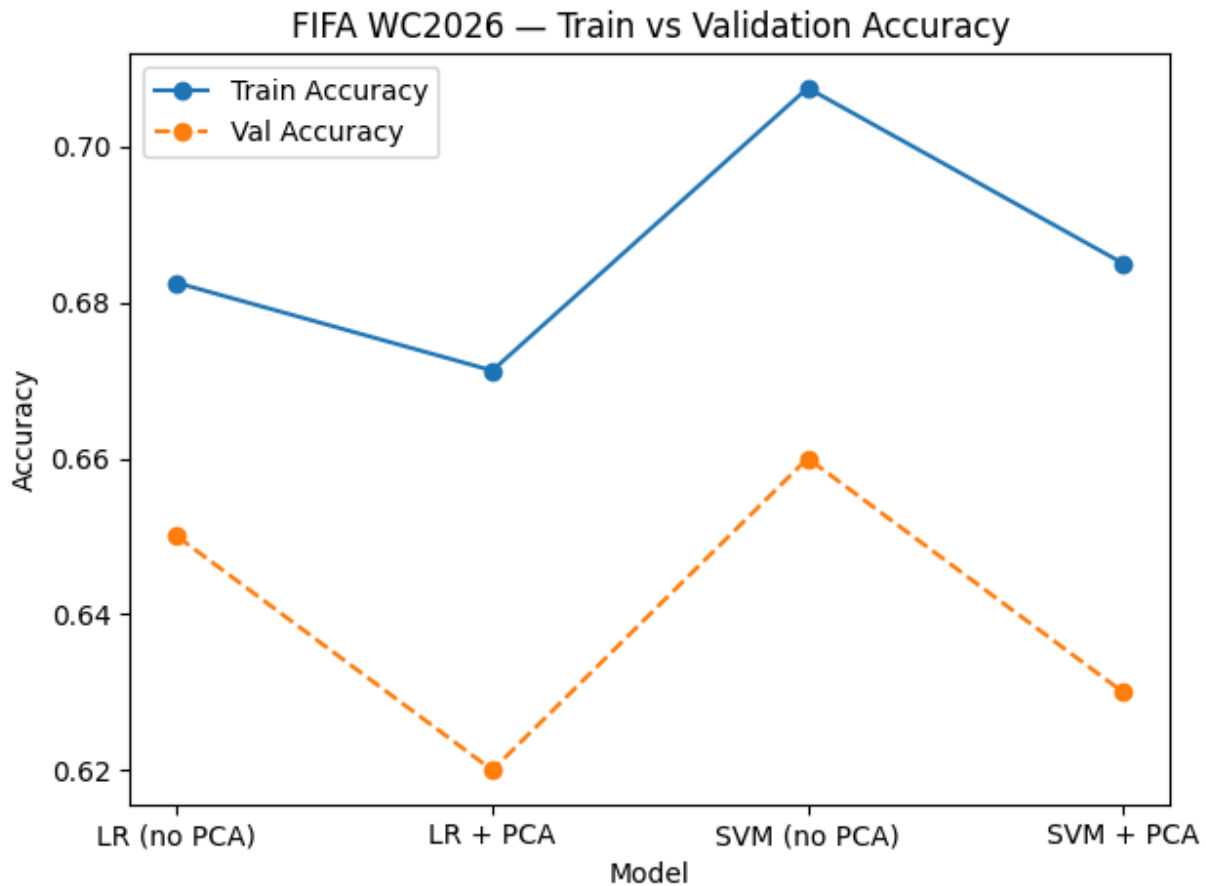


Figure 1. Train and validation accuracy across the four model configurations.

The train and validation accuracy plot shows that the SVM without PCA had the strongest validation performance. The training accuracy was higher than the validation for all models, which is expected because each model is evaluated on the same data it was trained on when calculating training accuracy. However, the gap was not extremely large, so the models did not appear to be severely overfitting.

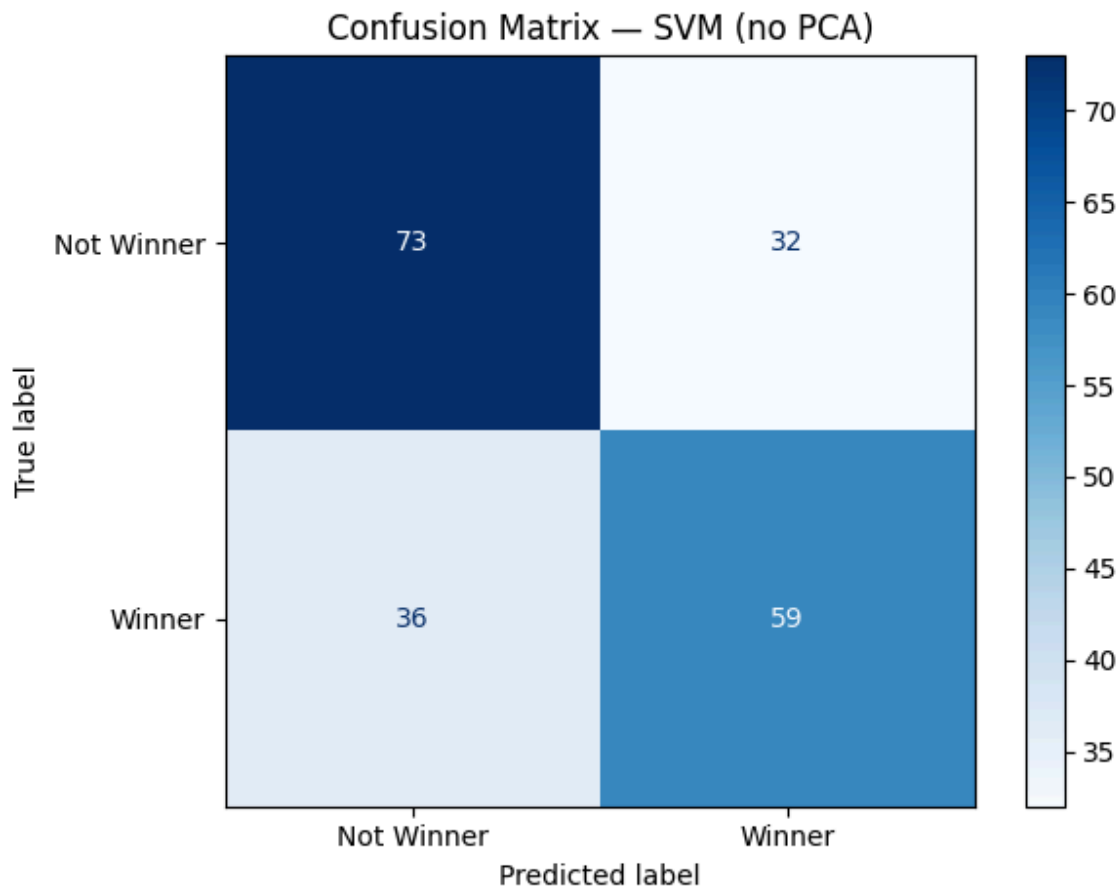


Figure 2. Confusion matrix for the SVM with RBF kernel without PCA.

The confusion matrix for the best model gives a better look at the SVM's predictions. Out of 105 non-winner examples, the model correctly classified 73 and incorrectly classified 32 as winners. Out of 95 winner examples, the model correctly classified 59 and incorrectly classified 36 as non-winners. This shows that the model performed slightly better than on the non-winner class, but its performance across both classes was fairly balanced.

7 Discussion

The results show that the SVM with an RBF kernel performed slightly better than Logistic Regression, but the difference was small. The SVM without PCA achieved the highest validation accuracy and F1-score, which suggests that some nonlinear relationships may exist in the dataset. This makes sense for soccer predictions because outcomes are rarely determined by one statistic alone. Ranking, recent form, scoring ability, and defensive strength may interact in ways that a linear model cannot fully capture.

At the same time, Logistic Regression remained competitive. Its F1-score was only slightly below the SVM's, so the nonlinear model was not a major improvement over the baseline. This is important because a more complex model is not automatically better. In this case, the SVM produced the best result, but Logistic Regression still captured much of the useful information in the feature set.

The PCA results were also useful. Both models performed worse with PCA than without it. This suggests that the original 22 features were already manageable and that reducing them did not help classification. Although PCA preserved 95% of the variance, it may have compressed information that was useful for predicting the winner label. Since PCA does not use the target variable, it can keep variation that is mathematically important but not necessarily predictive.

Overall, the best validation accuracy was 0.6600. This is better than random guessing, but it is not strong enough to treat the model as a reliable real-world predictor. Soccer outcomes are affected by many factors that are difficult to capture in a static dataset, such as injuries, lineups, coaching decisions, weather, and random events during a match. The results show that machine learning can find some useful patterns in the data, but the prediction problem remains difficult.

8 Limitations and Future Work

One limitation of this project is the size of the labeled dataset. The training file contained 1000 labeled examples, and only 200 examples were used for validation. With a dataset of this size, the results may depend partly on the specific train-validation split.

Another limitation is that the dataset uses team-level statistics rather than direct match-level comparisons. A match-level dataset would likely be more realistic because it could compare two teams against each other in a specific game. For example, instead of predicting a label from one team's statistics alone, the model could use the difference between the two teams' rankings, recent form, scoring rates, and defensive records.

For future work, hyperparameter tuning could also improve performance, especially for the SVM parameters. Additional models could be tested as well. Gradient Boosting or Neural Networks might be able to capture patterns that Logistic Regression and SVM did not, but these models would need to be compared using the same validation process.

9 Conclusion

This project compared Logistic Regression and Support Vector Machine models for predicting FIFA World Cup 2026 outcome labels from team-level soccer statistics. Each model was tested with and without PCA to evaluate both model choice and dimensionality reduction.

The best-performing model was the SVM with an RBF kernel without PCA. It reached a validation accuracy of 0.6600 and an F1-score of 0.6344. Logistic regression without PCA was

close behind, showing that the simpler baseline still performed reasonably well. PCA did not improve performance for either model, suggesting that the original feature representation was more useful for this task.

Overall, the project shows that machine learning can identify patterns in team-level soccer data, but it also highlights the limits of sports prediction. The model performed better than guessing, but the results were not strong enough to guarantee reliable predictions. A larger dataset, match-level features, and hyperparameter tuning would likely improve future versions of the project.

References

Cortes C., and Vapnik, V. (1995). Support-vector networks. Machine Learning, <https://link.springer.com/article/10.1007/BF00994018>

Jolliffe, I. T. (2002). Principal Component Analysis (2nd ed.). Springer. <https://books.google.com/books?id=TtVF-ao4fl8C>

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research. <https://doi.org/10.48550/arXiv.1201.0490>

Rambe, R. F (2026). FIFA World Cup 2026 Prediction System. Kaggle. <https://www.kaggle.com/dsv/16165118>

Github Link

<https://github.com/VeniaminV/FIFA-WC2026-Outcome-Predictor>